IBM FileNet P8 V5.6.x

Step-by-Step Hands-On Lab Guide

Day 5 – Storage & Security Administration

Storage Configuration | FileNet Security Model | LDAP Integration with WebSphere VMM

Author	Course Day	Lab Count	Total Duration
Rajesh Pasham	**Day 5 of 10**	**4 Labs**	**~2.5 Hours**

LAB-05-01	LAB-05-02	LAB-05-03	LAB-05-04
Configure Storage Areas *45 min | File system & FCD storage*	**Configure Storage Policies** *30 min | Policy assignment & testing*	**Implement Document Security** *40 min | ACLs, groups & inheritance*	**Configure LDAP via VMM** *35 min | WebSphere VMM integration*

LAB-05-01 | Configure Storage Areas

Duration: 45 Minutes | Create and configure file system and FCD storage areas

NOTE: START STATE: Content Platform Engine 5.2.1 is installed. P8Domain exists. TrainingOS Object Store is configured. WAS FileNetServer profile is running on port 9080. ACCE is accessible at http://localhost:9080/acce.

Objective

In this lab you will:

- Create a file system directory to serve as the storage area root on the local file system
- Create a File System Storage Area (FS_TrainingStore_01) in ACCE linked to TrainingOS
- Verify the storage area is Online and test by uploading a document
- (Optional Extension) Create a second storage area for a tiered storage simulation

STEP 1 | Create the Storage Area Directory on the File System | 5 min

Before creating the storage area in ACCE, the root directory must exist on the file system. FileNet will not create this directory automatically.

1. Open a Command Prompt as Administrator on the FileNet server
2. Create the storage area root directory:

Command	`mkdir C:\FileNetStore\TrainingOS\FS01`
Purpose	`Root directory for the primary file system storage area`

3. Also create a second directory for the optional extension lab:
   `mkdir C:\FileNetStore\TrainingOS\FS02`
4. Verify both directories exist:
   `dir C:\FileNetStore\TrainingOS\`

	VERIFICATION CHECKLIST
[ ]	Directory C:\FileNetStore\TrainingOS\FS01 exists
[ ]	Directory C:\FileNetStore\TrainingOS\FS02 exists

STEP 2 | Open ACCE and Navigate to Storage Areas | 3 min

5. Open a browser and go to: http://localhost:9080/acce
6. Log in with Username: ceadmin, Password: ceadmin
7. In the left navigation tree, expand P8Domain
8. Expand Object Stores
9. Expand TrainingOS
10. Click Storage (sub-menu will expand)
11. Click Storage Areas

[SCREENSHOT]
ACCE left navigation tree showing TrainingOS > Storage > Storage Areas node selected

STEP 3 — Create the File System Storage Area — 10 min

12. In the Storage Areas panel (right side), click New Storage Area
13. The New Storage Area wizard opens. Fill in the following fields:

Storage Area Type	`File System Storage Area (select from dropdown)`
Symbolic Name	`FS_TrainingStore_01`
Display Name	`Training File Store 01`
Description	`Primary file system storage area for TrainingOS object store`
Root Directory Path	`C:\FileNetStore\TrainingOS\FS01`
Maximum Content Size	`(Leave blank - no limit)`
Accessibility	`Online`
Affinity Cost	`10`

IMPORTANT: The Symbolic Name cannot contain spaces. Use underscores. This name is used in storage policies and cannot be changed after creation.

14. Click OK to create the storage area
15. The storage area FS_TrainingStore_01 should now appear in the list

[SCREENSHOT]
ACCE Storage Areas panel showing FS_TrainingStore_01 with Online status

	VERIFICATION CHECKLIST
[ ]	Storage area FS_TrainingStore_01 appears in the ACCE list
[ ]	Status shows: Online
[ ]	Root Directory Path: C:\FileNetStore\TrainingOS\FS01
[ ]	Affinity Cost: 10

STEP 4 — Set as Default Storage Area — 5 min

16. In ACCE, navigate to Object Stores > TrainingOS > Properties
17. Click the Storage tab
18. In the Default Storage Area field, select FS_TrainingStore_01 from the dropdown
19. Click Save

NOTE: Setting a default storage area ensures that documents without an explicit storage policy assignment are stored in this area. In a production system, you would typically set this at the object store level and override at the document class level for different tiers.

STEP 5 — Verify Storage by Uploading a Test Document (12 min)

20. Navigate to Object Stores > TrainingOS > Browse
21. Right-click the Root Folder > New Document
22. Fill in document properties:

Document Class	`Document (generic)`
Document Title	`Storage Test Document 01`
Filename	`StorageTest01.txt (create a simple text file first)`

23. To create a simple test file, open Notepad and save as: C:\Temp\StorageTest01.txt with content: FileNet Storage Area Test
24. In the Add Document wizard, click Browse and select C:\Temp\StorageTest01.txt
25. Click Next through all wizard steps
26. Click Finish to store the document
27. Right-click the newly created document > Properties
28. Select the Info tab and look for the Storage Area field

	VERIFICATION CHECKLIST
[ ]	Document StorageTest01.txt created successfully in TrainingOS Root Folder
[ ]	Document Properties > Info tab shows Storage Area: FS_TrainingStore_01
[ ]	Verify file physically stored: check C:\FileNetStore\TrainingOS\FS01 - a subfolder should be created

STEP 6 — (Extension) Create Second Storage Area for Tier 2 (10 min)

TIP: This extension step creates a second storage area with higher affinity cost to simulate a storage tier for migrated content. It will be used in Lab-05-02 when setting up a storage policy.

29. Repeat Steps 2-3 with the following values for the second storage area:

Storage Area Type	`File System Storage Area`
Symbolic Name	`FS_TrainingStore_02`
Display Name	`Training File Store 02 (Archive Tier)`
Root Directory Path	`C:\FileNetStore\TrainingOS\FS02`
Accessibility	`Online`
Affinity Cost	`50`

	VERIFICATION CHECKLIST
[ ]	Storage area FS_TrainingStore_02 appears in ACCE list with Online status
[ ]	Affinity Cost for FS_TrainingStore_02 is 50 (higher than FS_TrainingStore_01 = 10)

LAB-05-02 | Configure Storage Policies

Duration: 30 Minutes | Create and assign a storage policy to the TrainingOS object store

NOTE: START STATE: Lab-05-01 completed. Storage areas FS_TrainingStore_01 and FS_TrainingStore_02 exist in Online status.

Objective

In this lab you will:

- Create a storage policy STD_Policy_TrainingOS that places new documents in FS_TrainingStore_01
- Assign the storage policy as the default for the TrainingOS object store
- Create a document class-level storage policy override for a high-priority class
- Test that new documents are routed to the correct storage area per policy

STEP 1 | Create the Storage Policy | 12 in

1. In ACCE, navigate to Object Stores > TrainingOS > Storage > Storage Policies
2. Click New Storage Policy
3. Fill in the policy wizard fields:

Policy Name	`STD_Policy_TrainingOS`
Display Name	`Standard Storage Policy - TrainingOS`
Initial Storage Area	`FS_TrainingStore_01`
Destination Storage Area	`FS_TrainingStore_02 (archive after 90 days)`
Trigger Type	`Time After Document Creation`
Trigger Value (days)	`90`

4. Click Next and review the summary
5. Click Finish to create the policy

[SCREENSHOT]
ACCE Storage Policies panel showing STD_Policy_TrainingOS with Initial Area and Destination Area columns

STEP 2 | Assign Storage Policy as Object Store Default | 8 min

6. Navigate to Object Stores > TrainingOS > Properties
7. Click the Storage tab
8. In the Default Storage Policy field, select STD_Policy_TrainingOS
9. Click Save

NOTE: The default storage policy applies to all document classes in TrainingOS unless overridden at the class level. Existing documents are not affected - only new documents created after this assignment.

STEP 3	Test Document Storage Against Policy	10 min

10. Navigate to Object Stores > TrainingOS > Browse > Root Folder
11. Create a new test document:

Document Title	`Storage Policy Test 01`
Content File	`C:\Temp\PolicyTest01.txt (create in Notepad)`

12. After creation, right-click the document > Properties > Info tab
13. Verify the Storage Area field shows FS_TrainingStore_01
14. Verify the Storage Policy field shows STD_Policy_TrainingOS

	VERIFICATION CHECKLIST
[ ]	New document Storage Policy Test 01 created successfully
[ ]	Document Properties > Info > Storage Area: FS_TrainingStore_01
[ ]	Document Properties > Info > Storage Policy: STD_Policy_TrainingOS
[ ]	File visible on file system at C:\FileNetStore\TrainingOS\FS01\

LAB-05-03 | Implement Document Security

Duration: 40 Minutes | Configure ACLs, security inheritance, and role-based access

NOTE: START STATE: Lab-05-01 and Lab-05-02 completed. TrainingOS has content in FS_TrainingStore_01. User ceadmin is logged in to ACCE.

Objective

In this lab you will:

- View and understand the default ACL of the TrainingOS object store root folder
- Create a restricted subfolder with an explicit ACL that limits access to a specific group
- Verify that a non-admin user respects ACL restrictions
- Configure security inheritance settings on a document class

STEP 1 — Review the Root Folder Default ACL (8 min)

1. In ACCE, navigate to Object Stores > TrainingOS > Browse
2. Right-click Root Folder > Properties
3. Click the Security tab
4. Review the existing ACEs. Note the following:

Expected Entries	`#AUTHENTICATED-USERS with View Content + Read permissions`
Expected Entries	`ceadmin with Full Control (Change Security)`
Expected Entries	`#CREATOR-OWNER with Write + Major Version + Minor Version`

[SCREENSHOT]
ACCE Security tab showing Root Folder ACL with ACE entries for authenticated users and ceadmin

5. Record the existing ACEs in your lab notebook before making any changes

STEP 2 — Create a Restricted HR Documents Folder (5 min)

6. In ACCE Browse panel, right-click Root Folder > New Folder
7. Fill in the folder properties:

Folder Name	`HR_Confidential`
Folder Class	`Folder (generic)`
Parent Folder	`Root Folder (default)`

8. Click Finish to create the folder

STEP 3 — Configure Restricted ACL on HR_Confidential Folder — 12 min

9. Right-click the HR_Confidential folder > Properties
10. Click the Security tab
11. Click the Remove All Inherited Entries button to clear inherited ACEs

> **WARNING:** Removing inherited entries isolates this folder from parent ACL changes. Make sure to add back administrative access before saving, or you may lock yourself out.

12. Click Add to add a new ACE for the admin:

Principal	`ceadmin`
Access Type	`Allow`
Permissions	`Full Control (all permissions)`

13. Click Add again to add access for the HR group. In the training lab, use #AUTHENTICATED-USERS as a stand-in for an HR LDAP group:

Principal	`#AUTHENTICATED-USERS`
Access Type	`Allow`
Permissions	`View Content, Read, Write, Create Child, Minor Version`

14. The final ACL should have exactly 2 ACEs: ceadmin (Full Control) and #AUTHENTICATED-USERS (limited)
15. Click Save to apply the ACL

[SCREENSHOT]
ACCE Security tab for HR_Confidential folder showing 2 ACE entries

STEP 4 — Test ACL by Adding Document to Restricted Folder — 8 min

16. Navigate to Browse > Root Folder > HR_Confidential
17. Right-click > New Document

Document Title	`HR Confidential Policy 2025`
Content File	`C:\Temp\HR_Policy.txt (create in Notepad)`

18. Click Finish to add the document
19. Right-click the document > Properties > Security tab
20. Observe that the document inherited its ACL from the HR_Confidential folder

VERIFICATION CHECKLIST

[ ]	HR_Confidential folder created in Root Folder
[ ]	Folder ACL has exactly 2 ACEs: ceadmin (Full Control) + #AUTHENTICATED-USERS (limited)
[ ]	HR Confidential Policy 2025 document created in HR_Confidential folder
[ ]	Document ACL inherited from parent folder - same 2 ACEs visible

STEP 5 Configure Document Class-Level Security (7 min)

21. In ACCE, navigate to Object Stores > TrainingOS > Data Design > Document Classes
22. Click on Document (the generic base class)
23. Click the Security tab
24. Review the class-level ACL
25. Click Add to add a new ACE for group-level access:

Principal	`#AUTHENTICATED-USERS`
Access Type	`Allow`
Permissions	`View Content, Read (read-only access to all documents of this class)`

26. Click Save

> **TIP:** Document class ACLs apply to all new instances of that class regardless of where they are stored. Use class-level security to set baseline permissions, then override at the folder level for more restrictive areas like HR_Confidential.

	VERIFICATION CHECKLIST
[ ]	Document class ACL updated with #AUTHENTICATED-USERS having View Content + Read
[ ]	Setting saved successfully in ACCE

LAB-05-04 | Configure LDAP via WebSphere VMM

Duration: 35 Minutes | Configure VMM to connect to LDAP and verify user/group resolution

NOTE: START STATE: WAS FileNetServer profile is running. The training environment uses WAS embedded LDAP (which is already configured in the default VMM repository). This lab verifies and extends the existing configuration to understand the full VMM LDAP workflow.

Objective

In this lab you will:

- Verify the current VMM federated repository configuration in WAS Admin Console
- Review and confirm the LDAP repository connection settings for the training environment
- Test user lookup via the VMM repository using the WAS Admin Console user search
- Assign a FileNet role to an LDAP group and verify access in ACCE
- Configure and test LTPA token settings for single sign-on

STEP 1 | Access WAS Admin Console and Navigate to Federated Repositories | 5 min

1. Open a browser and navigate to: https://localhost:9043/ibm/console
2. Accept the self-signed certificate warning (click Advanced > Proceed)
3. Log in with:

Username	`wasadmin`
Password	`wasadmin`

4. In the left navigation panel, click Security
5. Click Global Security
6. In the User account repository section, verify Federated repositories is selected
7. Click the Configure button next to Federated repositories

[SCREENSHOT]
WAS Admin Console > Security > Global Security showing Federated repositories selected and Configure button

STEP 2 | Review the LDAP Repository Configuration | 8 min

8. On the Federated Repositories page, note the realm name: defaultWIMFileBasedRealm
9. Under Repositories in the realm, you should see entries for the file-based repository
10. Click Manage Repositories to see all configured repositories
11. Identify the existing LDAP repository entry (or file-based repository if LDAP is not pre-configured)

NOTE: In the training lab, WAS uses a file-based repository (defaultWIMFileBasedRealm) which acts as the user store. In a production environment, you would add an LDAP repository here. The steps below show how to ADD a new LDAP repository connection.

12. Click Add > LDAP repository to see the LDAP configuration form. Fill in these training values:

Repository Identifier	`TrainingLDAP`
Primary host	`localhost`
Port	`389`
Bind DN	`uid=wpsbind,o=defaultWIMFileBasedRealm`
Bind password	`(as configured - check with instructor)`
Base entry DN	`o=defaultWIMFileBasedRealm`
LDAP server type	`Custom (or IBM Tivoli Directory Server)`

13. Click Test Connection to verify LDAP connectivity

[SCREENSHOT]
WAS Admin Console LDAP repository configuration page with Test Connection button result showing Success

STEP 3 — Verify User Lookup via VMM (8 min)

14. In WAS Admin Console, navigate to: Users and Groups > Manage Users
15. In the search field, type: ceadmin and click Search
16. The user ceadmin should appear in the results
17. Click the ceadmin entry to view user details and group memberships

Expected User ID	`ceadmin`
Expected Display Name	`Content Engine Admin (or ceadmin)`
Expected Groups	`FileNet_Administrators (or as configured)`

18. Also search for: p8user and verify training users exist
19. Navigate to Users and Groups > Manage Groups
20. Search for: filenet and note all FileNet-related groups

	VERIFICATION CHECKLIST
[ ]	User ceadmin found in VMM user search with correct display name
[ ]	User ceadmin is a member of at least one FileNet group
[ ]	Group search for filenet returns at least one result

STEP 4 — Assign FileNet Admin Role to LDAP Group in ACCE (10 min)

21. Switch back to ACCE: http://localhost:9080/acce (log in as ceadmin if session expired)
22. Navigate to Object Stores > TrainingOS > Properties

23. Click the Security tab to view the Object Store-level ACL
24. Click Add to add a new group-level ACE:

Principal (search)	`filenet_admins (or the admin group found in Step 3)`
Access Type	`Allow`
Permissions	`Full Control`

25. If the group is not found, try searching with the full DN: cn=filenet_admins,o=defaultWIMFileBasedRealm
26. Click OK to add the ACE, then Save

> **TIP:** In the training lab, you may need to use the exact VMM group name format. If user/group search in ACCE returns no results, verify the VMM configuration in WAS Admin Console and check the WAS SystemOut.log for LDAP search errors.

	VERIFICATION CHECKLIST
[ ]	Group-level ACE added to TrainingOS object store for filenet_admins group
[ ]	Save completed without error
[ ]	Object store ACL now includes both ceadmin (explicit) and the group-level entry

STEP 5 — Verify and Configure LTPA Token Settings (4 min)

27. In WAS Admin Console, navigate to: Security > Global Security
28. Scroll to the Authentication section, click LTPA
29. Review and record the current LTPA settings:

Token Timeout (minutes)	`Default: 30. Record current value.`
Allow propagation	`Should be checked (enabled)`
Interoperability	`LTPA version 2 (default in WAS 8.5+)`

30. Change the Token Timeout to 60 minutes for the training lab (reduces repeated logins)
31. Click OK and Save to master configuration
32. Note the message: 'A server restart may be required'. For this lab, continue without restart.

	VERIFICATION CHECKLIST
[ ]	LTPA settings page accessible in WAS Admin Console
[ ]	Token Timeout changed to 60 minutes and saved to master configuration
[ ]	No error messages during save

Day 5 Lab Completion Summary

Lab	Title	Skills Practiced	Status
LAB-05-01	Configure Storage Areas	File system, ACCE storage	**[ ] Complete**
LAB-05-02	Configure Storage Policies	Policy creation, assignment	**[ ] Complete**
LAB-05-03	Implement Document Security	ACLs, inheritance, groups	**[ ] Complete**
LAB-05-04	Configure LDAP via VMM	WAS VMM, LTPA, user search	**[ ] Complete**